

WASP-35b

R-1372

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-35_220215 · created 2026-08-01T00:56:18+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459625.7162 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.12 +/- 0.021
Transit depth (Rp/Rs)^2	1.43 +/- 0.49 [%]
Orbital Inclination (inc)	89.81 +/- 1.28
Airmass coefficient 1 (a1)	1.391 +/- 0.022
Airmass coefficient 2 (a2)	-0.1208 +/- 0.0025
Scatter in the residuals of the lightcurve fit is	1.43 %
Best Comparison Star	None
Optimal Aperture	4.64
Optimal Annulus	12.07
Transit Duration (day)	0.1336 +/- 0.0037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.12 $\pm$ 0.021 vs 0.1241 (deviation 0.13 σ)
Duration fit vs published	0.1336 d vs 0.1317 d (deviation 0.34 σ)
Mid-time O–C	7.3 min
Depth SNR	3.7
Residual scatter	1.43 %

Light curve

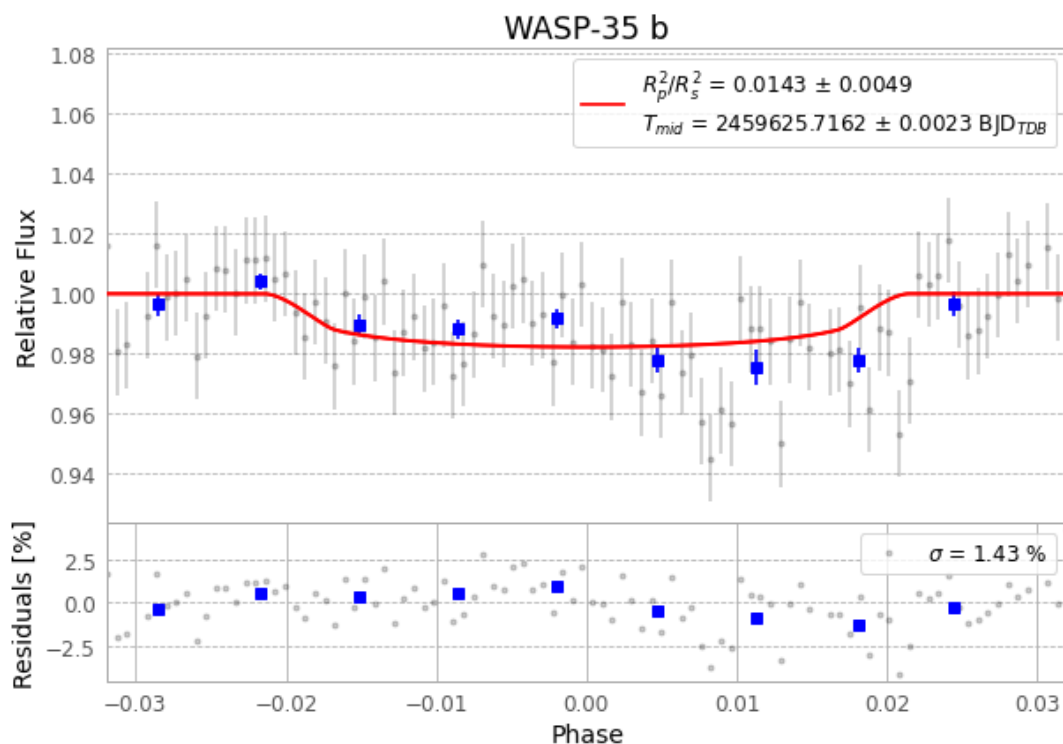


Figure 1 — FinalLightCurve_WASP-35 b_2022-02-14.png (SHA-256 a143ccd1e1afa973...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [218, 241], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e856043eb7f6c1de...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

98 FITS frames (64 MB), WASP-35220215024212.FITS → WASP-35220215073312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-220215.log (fe2207248ee3...), FinalLightCurve_WASP-35 b_2022-02-14.png (a143ccd1e1af...), FinalLightCurve_WASP-35 b_2022-02-14.csv (d334ce6ce939...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 00:56Z.